

# Does the Pareto-optimal RAG configuration transfer across GPU tiers?

An experiment report

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August 3, 2026

## Abstract

Papers that search for good retrieval-augmented generation (RAG) configurations publish a recommended setup benchmarked on the authors' hardware. Practitioners copy it onto cheaper cards. This report tests whether the recommendation survives that move. I ran an identical 96-configuration grid — chunk size, overlap, embedding model, top- $k$ , reranker — on an NVIDIA A100-SXM4-40GB and an NVIDIA L4, against one corpus and one hand-labelled gold set, with every result record provenance-stamped. Retrieval quality is deterministic and came out identical on both cards, so every change in the result is attributable to latency alone. The Pareto frontier is not the same on both: four configurations on the A100, six on the L4, three shared, with Kendall's  $\tau = 0.802$  between the two latency rankings. The difference survives both latency statistics and an  $\varepsilon$ -dominance tolerance of 300ms. My pre-registered hypothesis — that cross-encoder reranking would flip from optimal to dominated on the weaker card — did not hold: reranking appears on both frontiers. The practical consequence is that copying a published shortlist onto different hardware can leave you running a configuration that is slower than an equally accurate alternative, at no visible cost in quality.

## 1 Introduction

Every paper that searches the RAG configuration space ends with a recommendation: this chunk size, this top- $k$ , rerank or do not. That recommendation is produced once, on whatever GPU the authors had, and then travels — into blog posts, into default settings, into production systems running on hardware an order of magnitude weaker. The move is never validated, because the experiment that would validate it requires running the same grid twice on deliberately mismatched hardware, and configuration-search papers are built to explore the algorithm space rather than the hardware axis. The result is a literature of optimality claims with an unstated precondition attached.

This report supplies the missing measurement. **The same 96-configuration grid, the same corpus, the same hand-labelled gold set and the same code were executed on two GPU tiers separated by roughly  $5.2\times$  in memory bandwidth**, and both result sets were recorded in full. Three properties make the comparison worth trusting:

- **The hypothesis and its refutation condition were committed to version control before the first cell ran**, and neither was edited afterwards. The result reported here is the one the experiment was designed to be able to contradict.
- **Retrieval quality came out bit-identical across cards**, which removes the confound entirely: every difference in the result is a latency effect, and nothing else is available to explain it.
- **Every record is self-describing** — GPU, driver, library versions, git SHA, corpus and gold-set hashes — so no claim here depends on external state, and any cell can be regenerated from its identifier alone.

The question is: **does the Pareto-optimal RAG configuration transfer across GPU tiers, or does the ranking of configurations change?**

**Hypothesis, registered in advance.** The Pareto-optimal configuration is not hardware-invariant. Chunk size and top- $k$  keep their optimal settings, but cross-encoder reranking flips from Pareto-optimal on the A100 to Pareto-dominated on the L4, because its latency cost grows faster than the recall it buys.

**Refutation condition.** Both frontiers contain the same configurations in the same order.

**Outcome.** The frontiers differ, and the difference is robust to both the choice of latency statistic and to 300ms of assumed host noise. The mechanism named in the hypothesis did not occur: reranking holds a frontier slot on both cards. §4 reports which claim the evidence supports and which it does not, and §5 states what would be needed to settle the remainder.

## 2 Related work

Multi-objective RAG configuration search is a crowded literature, and this report does not claim to be first to locate a Pareto frontier over retrieval configurations. **syftr** [2] runs multi-objective Bayesian optimisation over agentic and non-agentic RAG flows, balancing accuracy against cost on a single hardware target. **RAG-Stack** [4] identifies the quality–performance frontier across the joint algorithm–system design space, and is the closest prior work to acknowledge hardware at all — but it explores from the algorithm side, treating the system as a fixed backdrop. **RAGSmith** [5] runs evolutionary search over complete pipelines and reports sensitivity to the dataset it was tuned on, which is adjacent to but distinct from hardware sensitivity. **METIS** [9] performs joint algorithm/system exploration, again on fixed hardware. The underlying RAG approach is due to Lewis et al. [7].

In all four, the hardware tier is a constant rather than a variable. None asks whether the configuration it recommends would still be recommended on a smaller GPU.

*Existing work locates the configuration frontier on fixed hardware; whether the frontier’s composition is stable across hardware tiers has not been tested.*

The finding has the same shape as an earlier result of my own [11], which found the optimal RAG *strategy* depends on model scale where this one finds the optimal *configuration* depends on hardware tier. Both say the same thing about what “optimal” means in this literature: it is conditional on something the recommending paper held fixed.

## 3 Experimental setup

**Pipeline.** Recursive character chunking → **sentence-transformers** embedding [10] → FAISS [3] **IndexFlatIP** retrieval over  $L_2$ -normalised vectors → optional cross-encoder rerank → generation. Two BGE embedding models [12] are registered, **small** (384d) and **base** (768d), same family so the contrast is dimension rather than architecture. The reranker is **cross-encoder/ms-marco-MiniLM-L6-v2** — a Sentence-BERT cross-encoder [10] fine-tuned on the MS MARCO passage ranking data [1] — chosen as the lighter of two candidates specifically because it makes the hypothesis harder to confirm. The generator is **Qwen/Qwen2.5-3B-Instruct** [8], fixed and unswept: this is a retrieval-configuration experiment, not a generation-quality one. Greedy decoding, **max\_new\_tokens=128**, seed 1337, and an answer-only-from-context prompt (Appendix A).

**Corpus and gold set.** 15 PDFs on GenAI/LLM-engineering topics, 72 pages, 35,751 tokens, single-subject by design because that retrieves harder than 15 unrelated documents. 30 questions hand-labelled by one annotator, with labels recorded as **character spans in the source document**, never chunk IDs — chunk IDs exist only relative to one chunk size, and three of the five grid knobs change the chunking. A retrieved chunk counts as relevant if it overlaps a gold span by  $\geq 50\%$  of that span. Construction detail, including the planned question-type distribution and a lexical-saturation problem caught and fixed during review, is in Appendix B.

**Grid.** The first three knobs determine the index; the last two are query-time only, so 96 cells cost 12 index builds.

**Table 1:** The 96-cell grid.  $3 \times 2 \times 2 = 12$  indices;  $12 \times 4 \times 2 = 96$  cells per card; 192 runs total.

| Knob            | Levels                     | Count |
|-----------------|----------------------------|-------|
| Chunk size      | 256 / 512 / 1024 tokens    | 3     |
| Chunk overlap   | 0 / 15%                    | 2     |
| Embedding model | small (384d) / base (768d) | 2     |
| Top- $k$        | 3 / 5 / 10 / 20            | 4     |
| Reranker        | off / cross-encoder        | 2     |

**Measurement.** Four stages are timed separately — embed, search, rerank, generate — each with `torch.cuda.synchronize()` before the stop timestamp, since GPU kernels launch asynchronously and an unsynchronised stop measures CPU issue rate rather than GPU work. Three warmup queries are discarded, then each of the 30 gold queries runs three times;  $p_{50}$  and  $p_{95}$  are computed over all 90 measurements per cell rather than over per-query means.  $p_{95}$  rather than the mean because rented hosts have asymmetric tails. Every record carries GPU, driver, CUDA, library versions, git SHA, corpus and gold hashes, and (from the L4 run onward) model revisions. Full harness and provenance detail is in Appendix A.

**Metrics.** **recall@ $k$ :** 1 if any top- $k$  chunk is relevant to any gold span, averaged over 30 queries; reported at  $k = 5$  (headline) and  $k = 10$ . **MRR:** reciprocal rank of the first relevant chunk. **Context precision:** fraction of top-5 chunks that are relevant. All implemented as pure functions and unit-tested against boundary cases including the recall@ $k$  off-by-one at exactly rank  $k$ .

**Hardware.** A100-SXM4-40GB (Ampere, SM80, 40 GB,  $\sim 1,555$  GB/s) and L4 (Ada, SM89, 24 GB,  $\sim 300$  GB/s) — a bandwidth ratio of roughly  $5.2\times$ . The experiment was planned around an A100 80GB ( $\sim 2,039$  GB/s); the card actually provisioned was the 40GB SKU, confirmed from the provenance stamp rather than assumed. Software versions per card, and the reason the two sweeps carry different git SHAs, are in Appendix A.

## 4 Results

**Retrieval quality is identical across cards.** All 96 recall@5 values match exactly between the A100 and L4 records. Retrieval is deterministic given the same index, weights and seed, so the GPU changes speed and nothing else. Every result below is therefore a latency effect, with no confound from quality.

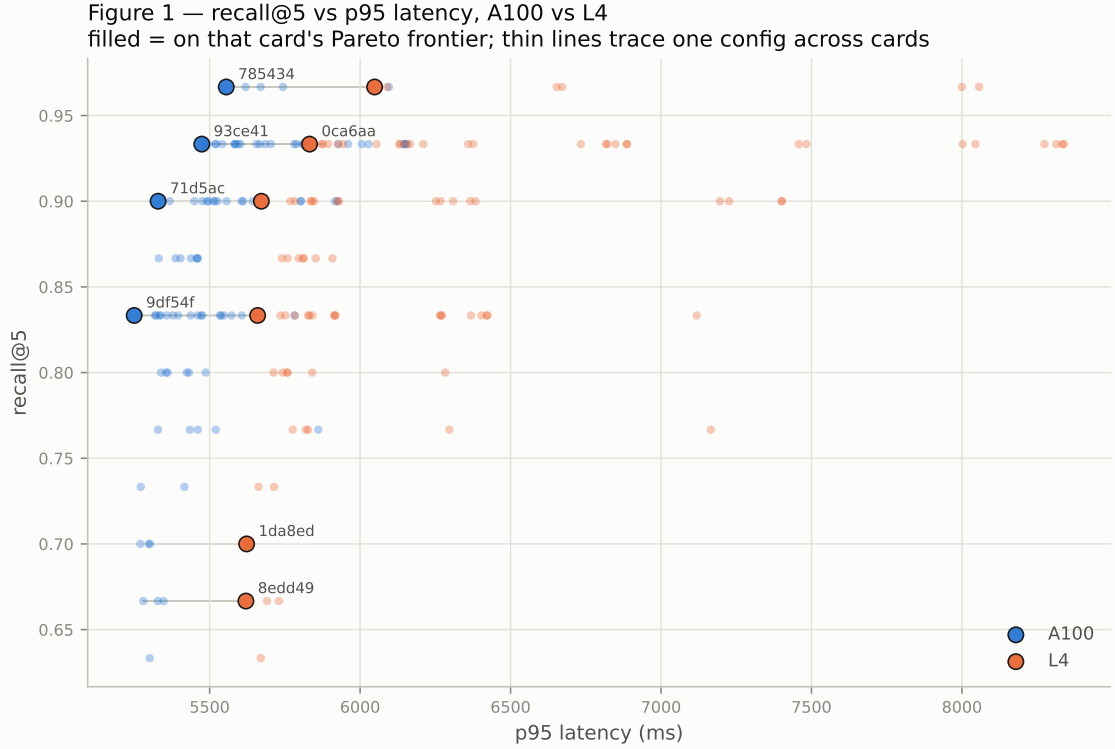
**The frontiers differ.** Seven configurations sit on at least one card’s frontier; the other 89 are dominated on both. The A100 frontier has four members, the L4 six, and three are shared.

**Table 2:** Every configuration on at least one frontier. Recall is identical on both cards, so only latency distinguishes them.

| chunk | ov.  | embed        | $k$ | reranker      | recall@5 | A100 $p_{95}$ | L4 $p_{95}$ | frontier    |
|-------|------|--------------|-----|---------------|----------|---------------|-------------|-------------|
| 256   | 0.0  | base         | 3   | off           | 0.833    | 5249 ms       | 5659 ms     | both        |
| 256   | 0.0  | base         | 5   | off           | 0.900    | 5329 ms       | 5672 ms     | both        |
| 1024  | 0.0  | base         | 5   | off           | 0.967    | 5555 ms       | 6048 ms     | both        |
| 1024  | 0.15 | <b>small</b> | 3   | cross-encoder | 0.933    | 5474 ms       | 5876 ms     | <b>A100</b> |
| 1024  | 0.15 | <b>base</b>  | 3   | cross-encoder | 0.933    | 5483 ms       | 5832 ms     | <b>L4</b>   |
| 256   | 0.0  | small        | 3   | off           | 0.667    | 5280 ms       | 5621 ms     | L4          |
| 256   | 0.0  | small        | 3   | cross-encoder | 0.700    | 5298 ms       | 5623 ms     | L4          |

Kendall’s  $\tau$  [6] between the two cards’ latency orderings is 0.802 over  $n = 96$ : most of the speed ranking transfers, with enough reordering to produce the difference above. No configuration ran out of memory on either card, contrary to expectation at `chunk=1024` $\times$ `top_k=20` $\times$ cross-encoder.

**The reranker did not flip.** Cross-encoder reranking appears on both frontiers, so the pre-registered mechanism did not occur. Its predicted economics are nonetheless visible: the rerank stage’s latency



**Figure 1:** recall@5 against  $p_{95}$  latency for all 96 configurations on both cards. Ringed points are that card’s Pareto frontier; grey lines connect the same configuration across cards.

grows  $1.40\times$  from A100 to L4 (ratio of means over the 48 reranker-enabled cells;  $1.25\times$  by median of per-cell ratios), against  $1.00\times$  for search,  $1.04\times$  for embed and  $1.17\times$  for generate. It is the one stage whose cost degrades disproportionately on the weaker card. It does not decide anything here because generation dominates: the rerank stage is a median 0.81% of a cell’s  $p_{95}$  (range 0.32–2.16%), so the disproportionate slowdown has nowhere large enough to act. Reranking buys +0.018 recall@5 on average, identically on both cards.

**What moved instead.** Two configurations differ only in embedding model — rows four and five of Table 2, both at recall 0.933 — and which one holds the frontier slot swaps between cards. The margin is 9 ms on the A100 and 44 ms on the L4, under 1% of a  $\sim 5,800$  ms cell.

**How far that survives.** Because those margins are small, the result was re-derived under two perturbations requiring no new measurement (Table 3). Varying the latency statistic tests whether the effect is an artifact of the tail. Widening domination by  $\varepsilon$  milliseconds tests how much host noise it tolerates.

**Table 3:** Frontier composition under both latency statistics and under  $\varepsilon$ -dominance on  $p_{95}$ . Reproducible via `scripts/sensitivity.py`.

|                        | A100 | L4 | shared | identical? | embedding swap present? |
|------------------------|------|----|--------|------------|-------------------------|
| $p_{95}$               | 4    | 6  | 3      | no         | yes                     |
| $p_{50}$               | 6    | 7  | 4      | no         | <b>no</b>               |
| $\varepsilon = 25$ ms  | 10   | 9  | 6      | no         | —                       |
| $\varepsilon = 100$ ms | 29   | 26 | 21     | no         | —                       |
| $\varepsilon = 300$ ms | 73   | 52 | 52     | no         | —                       |

Two claims of different strength follow. **That the frontiers differ is robust:** it holds under both statistics and at every  $\varepsilon$  up to 300 ms, roughly seven times the largest deciding margin. **That the**

**embedding model is the knob responsible is not:** under  $p50$  the frontiers still differ, but through five other configurations varying in chunk size, top- $k$  and reranker. The single-knob story is a property of the tail statistic. I report it as an illustrative case of how a frontier slot changes hands, not as an established claim about which knob is hardware-sensitive.

## 5 Threats to validity

**No repeated-sweep variance estimate.** Each of the 192 cells ran once per card, and only aggregate statistics were retained, so a bootstrap interval cannot be recovered retrospectively. The  $\varepsilon$ -analysis above bounds the noise the broad result tolerates without estimating it, but the 9–44 ms embedding-model margin is exactly the scale a shared host’s noisy neighbours could produce. Repeating only the configurations that decide frontier membership, rather than the full grid, would settle it cheaply.

**Three of four A100 model revisions are inferred.** Revision pinning originally covered only the embedding models, and wrote the pin to a file on whichever machine loaded first — a cloud VM, deleted after the sweep. The `small` revision survives because its pin file is git-tracked and was committed before the sweep; `base`, the reranker and the generator were reconstructed by resolving `main` afterwards, valid under the assumption that each repository was unchanged since its last update (11 months to 2.4 years before the sweep) and not force-pushed. The reranker is one arm of the hypothesis, so this matters. The gap is closed from the L4 run onward.

**Single annotator, and a constructed query distribution.** One person built the corpus and wrote all 30 questions, with no second annotator and no agreement figure. The query set was also built to a planned distribution rather than sampled from usage: 8 of 30 written to be lexically confusable across documents, 6 requiring two documents. Both proportions exceed what natural query traffic would likely produce, and both favour configurations that rerank and retrieve at higher  $k$ . Absolute recall is therefore not comparable to other corpora; only the relative ordering within this experiment is.

**Lexical saturation was mitigated, not eliminated.** A question phrased in its own answer span’s vocabulary is retrievable at any configuration, which would collapse both frontiers for a reason unrelated to hardware. Five such questions were caught and rewritten (Appendix B), but that review had no independent check behind it.

**Two cards, one corpus, one embedding family.** The hardware axis has two points, so this shows that composition changed between these tiers, not how it changes with GPU class in general — a gradual shift and a threshold effect would look identical here. Both embedding models are `BAAI/bge-*-en-v1.5`, so the sensitivity measured is to embedding *size* within one family.

**Single seed.** Fixed at 1337 and not varied. Narrower than it appears: decoding is greedy and retrieval is seed-independent, so the three repeats measure timing rather than generation variance.

## 6 Conclusion

**Tested.** One RAG pipeline, 96 configurations, two GPU tiers, identical corpus, gold set, code and model weights.

**Found.** Frontier membership differs — four winners on the A100, six on the L4, three shared — and the difference survives both latency statistics and a 300 ms domination tolerance. The reranker, which the hypothesis named, did not change status.

**Consequence.** Do not assume a published RAG configuration shortlist transfers unchanged to cheaper hardware. The cost of getting it wrong is not accuracy — accuracy is identical — but latency you did not need to spend, on a configuration an equally accurate alternative would have beaten. That loss is invisible unless you measure it.

This experiment does not identify which knob to watch, and §4 is explicit that the candidate it found does not survive a change of statistic. The defensible advice is about margins rather than knobs: any near-tied pair on a frontier is a hardware-transfer risk, so re-run the top few candidates on the deployment card rather than trusting the single published winner.

## A Setup detail

### Prompt.

Answer the question using ONLY the context below. If the context does not contain the answer, reply exactly: "I don't know based on the provided context."

Context:  
{context}

Question: {question}

Answer:

**Chunking.** Separators are tried coarse-to-fine — paragraph break, line break, sentence, word, raw character — descending into a piece only while it is over budget. Overlap carries trailing atoms from one chunk into the next rather than re-slicing. Every chunk retains {doc\_id, char\_start, char\_end} relative to its document’s extracted text, which is what allows span-based gold labels to survive a change of chunk size.

| Role      | Model                  | Revision    | Provenance |
|-----------|------------------------|-------------|------------|
| small     | BAAI/bge-small-en-v1.5 | 5c38ec7c... | measured   |
| base      | BAAI/bge-base-en-v1.5  | a5beb1e3... | inferred   |
| reranker  | ms-marco-MiniLM-L6-v2  | c5ee24cb... | inferred   |
| generator | Qwen2.5-3B-Instruct    | aa8e7253... | inferred   |

**Model revisions.** “Measured” means the revision is recorded in a git-tracked file at the commit stamped on the sweep’s records. “Inferred” means it was reconstructed afterwards; see §5.

|                       | A100         | L4           |
|-----------------------|--------------|--------------|
| Driver                | 580.82.07    | 580.82.07    |
| CUDA                  | 12.8         | 13.0         |
| torch                 | 2.11.0+cu128 | 2.13.0+cu130 |
| sentence-transformers | 5.6.0        | 5.6.1        |
| FAISS                 | 1.14.3       | 1.14.3       |
| git_sha               | 04caf49      | 3761de1      |

**Software and hashes.** Both content hashes are identical on all 192 records:

|            |                                          |
|------------|------------------------------------------|
| corpus_sha | a2a9aecc051466d25ff4f3e48aaf2c69aba1a8aa |
| gold_sha   | 3afa042d8d9cd6784e8e1b049e4f4f6c1d45709c |

The two sweeps carry different git\_sha values because the revision-pinning fix landed between them. The difference is provenance capture only: every file the retrieval and generation path executes is byte-identical between the two commits except two import statements from a package restructure.

**Other harness details.** Peak VRAM is the high-water mark across a cell (reset\_peak\_memory\_stats then max\_memory\_allocated). An out-of-memory cell is caught and written with status="oom" and no cost or quality block rather than crashing the sweep or being skipped; such a cell cannot be a frontier member by construction, and is reported in a refusals table. Indices are cached on (chunk\_size,

overlap, embed\_model) so the sweep rebuilds 12 times rather than 96. Any single cell regenerates from its config\_id with one command.

## B Gold set construction

Question types were allocated before any question was written: 7 with single-sentence answers, 9 with paragraph answers, 8 lexically confusable across documents, 6 requiring two documents. That yields 30 questions and 38 gold spans, covering all 15 documents, with span lengths from 38 to 414 characters (median 193).

Spans were located against the same pypdf-extracted text the pipeline indexes, not the PDF’s visual layout, using a helper that returns exact offsets so none were hand-counted. Thirteen initially-quoted spans did not resolve on the first pass, because code listings extract with line numbers interleaved, identifiers extract letter-spaced, and some punctuation swallows adjacent spaces. All were re-resolved and verified before the gold set was hashed and frozen.

Two review passes were run. The first found two mislabelled confusable-document pairs, corrected, and five questions with high content-word overlap against their own gold span (47–62%; one reused six of the span’s terms in the span’s order). A question phrased in its answer’s vocabulary is retrievable by near-lexical match regardless of configuration — recall saturates across the grid, both cards produce identical frontiers, and the result reads as a transfer finding when it is really a dead instrument. All five were rewritten to describe the answer without its source terminology, targeting the same spans.

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